EXPERIMENT-01

FIND-S Algorithm: Dataset-Based Python Programs

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1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Problem 1: Enjoying Sport

Aim

To implement FIND-S using a sport-related dataset stored in a separate CSV file and identify the most specific hypothesis consistent with positive examples.

Procedure

1. Save the training examples in sport_data.csv with the target class in the last column.
2. Run the Python program from the same folder as the CSV file.
3. Read the CSV rows and initialize the hypothesis from the first positive example.
4. Ignore negative examples; replace differing attribute values with '?' for each later positive example.
5. Display the final hypothesis.

Dataset File / Input

File name: sport_data.csv

The CSV file contains a header row followed by these training examples:

Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Cool	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Light	Warm	Change	Yes

Code

```
import csv

filename = "sport_data.csv"

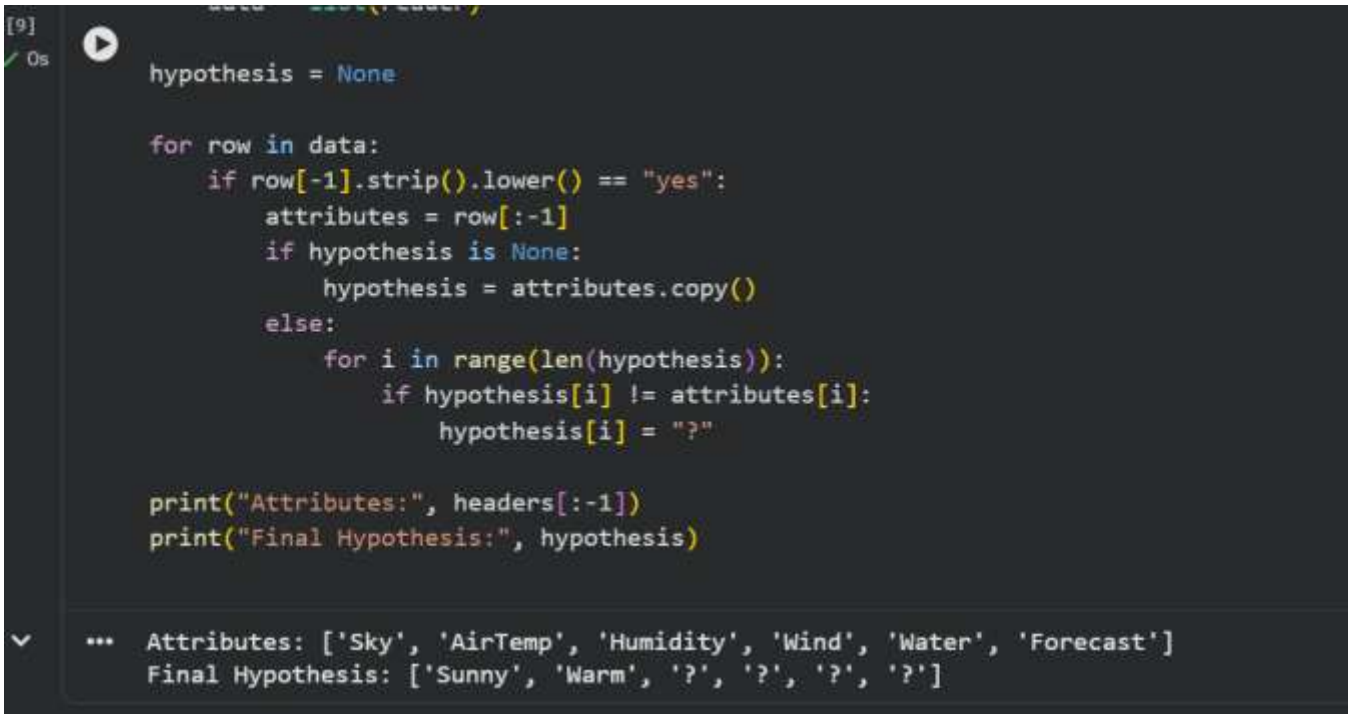
with open(filename, "r", newline="", encoding="utf-8") as file:
    reader = csv.reader(file)
    headers = next(reader)
    data = list(reader)

hypothesis = None

for row in data:
    if row[-1].strip().lower() == "yes":
        attributes = row[:-1]
        if hypothesis is None:
            hypothesis = attributes.copy()
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = "?"
```

```
print("Attributes:", headers[:-1])
print("Final Hypothesis:", hypothesis)
```

Output



Result

The final hypothesis keeps Sky = Sunny and AirTemp = Warm; the remaining attributes vary across positive examples.

Problem 2: Choosing a Laptop

Aim

To apply FIND-S to a laptop-selection dataset stored in a CSV file and determine which attributes remain common to positive selections.

Procedure

6. Create laptop_data.csv with laptop attributes and Selected as the target column.
7. Keep the CSV file and Python script in the same folder.
8. Read the dataset and initialize the hypothesis from the first row labelled Yes.
9. Skip rows labelled No and generalize differing attributes in positive rows to '?'.
10. Print the resulting hypothesis.

Dataset File / Input

File name: laptop_data.csv

The CSV file contains a header row followed by these training examples:

Budget	Weight	Battery	Purpose	Selected
High	Light	Long	Study	Yes
High	Heavy	Long	Gaming	Yes
Low	Light	Short	Study	No

High	Light	Short	Work	Yes

Code

```
import csv

filename = "laptop_data.csv"

with open(filename, "r", newline="", encoding="utf-8") as file:
    reader = csv.reader(file)
    headers = next(reader)
    data = list(reader)

hypothesis = None

for row in data:
    if row[-1].strip().lower() == "yes":
        attributes = row[:-1]
        if hypothesis is None:
            hypothesis = attributes.copy()
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = "?"

print("Attributes:", headers[:-1])
print("Final Hypothesis:", hypothesis)
```

Output



Result

Budget = High is the only attribute shared by all positive laptop selections.

Problem 3: Selecting a Planting Location

Aim

To demonstrate the FIND-S algorithm on a plant-growth dataset and find the most specific hypothesis from positive plant-growth examples.

Procedure

11. Store soil, sunlight, watering, season, and target labels in plant_data.csv.
12. Run the Python script from the directory containing the dataset.
13. Load each CSV row and use the first PlantGrows = Yes example to initialize the hypothesis.
14. Ignore No examples and replace attributes that differ across Yes examples with '?'.
15. Display the final hypothesis.

Dataset File / Input

File name: plant_data.csv

The CSV file contains a header row followed by these training examples:

Soil	Sunlight	Watering	Season	PlantGrows
Loamy	Full	Moderate	Spring	Yes
Loamy	Partial	Frequent	Summer	Yes
Sandy	Full	Low	Summer	No
Clay	Full	Moderate	Autumn	Yes

Code

```
import csv

filename = "plant_data.csv"

with open(filename, "r", newline="", encoding="utf-8") as file:
    reader = csv.reader(file)
    headers = next(reader)
    data = list(reader)

hypothesis = None

for row in data:
    if row[-1].strip().lower() == "yes":
        attributes = row[:-1]
        if hypothesis is None:
            hypothesis = attributes.copy()
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = "?"

print("Attributes:", headers[:-1])
print("Final Hypothesis:", hypothesis)
```

Output

```
[11]
✓ 0s
    for row in data:
        if row[-1].strip().lower() == "yes":
            attributes = row[:-1]
            if hypothesis is None:
                hypothesis = attributes.copy()
            else:
                for i in range(len(hypothesis)):
                    if hypothesis[i] != attributes[i]:
                        hypothesis[i] = "?"

    print("Attributes:", headers[:-1])
    print("Final Hypothesis:", hypothesis)

    Attributes: ['Budget', 'Weight', 'Battery', 'Purpose']
    Final Hypothesis: ['High', '?', '?', '?']

[12]
✓ 0s
    !mv /plant_data.csv /content/
```

Result

All four attributes differ among the positive examples, so none remains a fixed attribute in the final hypothesis.

Problem 4: Selecting a Travel Plan

Aim

To implement FIND-S for a travel-plan dataset stored separately and determine the common attribute values among positive trip decisions.

Procedure

16. Save the travel examples in travel_data.csv.
17. Place travel_data.csv beside problem4_travel.py.
18. Read the dataset and initialize the hypothesis with the first GoTrip = Yes example.
19. Ignore negative examples and generalize any mismatching attributes in positive examples to '?'.
20. Print the final hypothesis.

Dataset File / Input

File name: travel_data.csv

The CSV file contains a header row followed by these training examples:

Weather	Distance	Transport	Companions	GoTrip
Clear	Short	Train	Friends	Yes
Clear	Long	Car	Family	Yes
Stormy	Short	Bus	Alone	No
Cloudy	Short	Flight	Friends	Yes

Code

```
import csv

filename = "travel_data.csv"
```

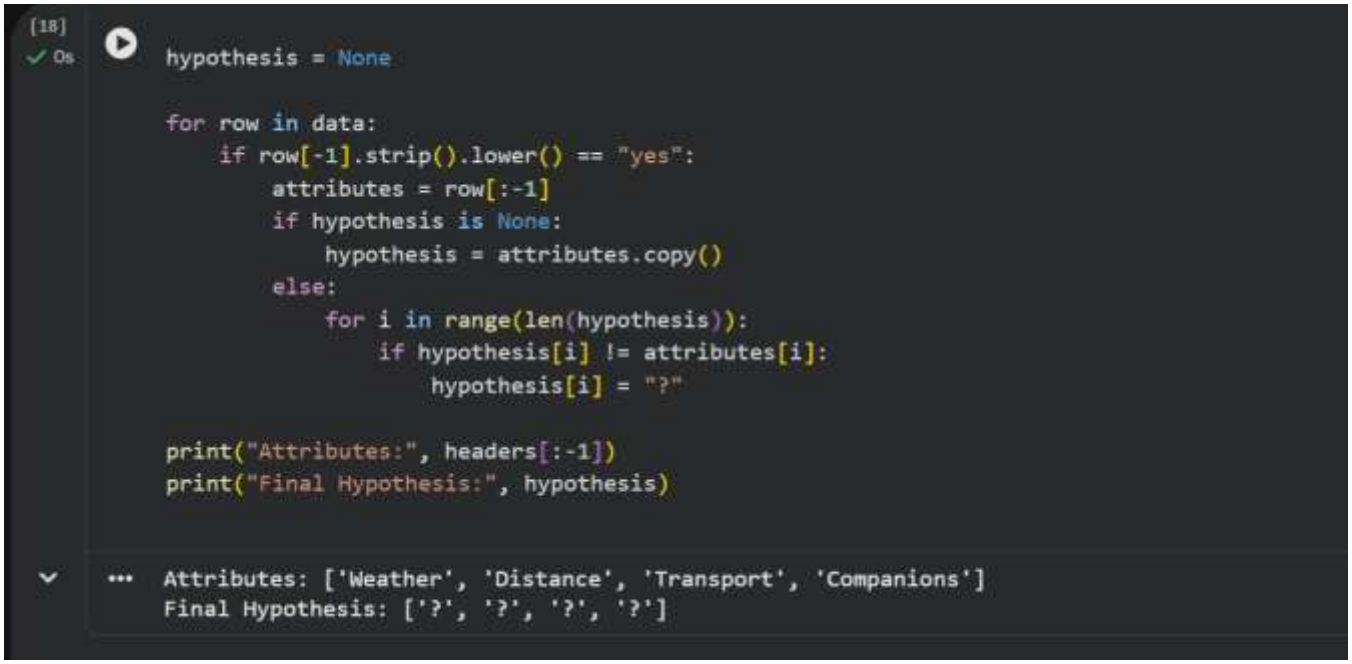
```
with open(filename, "r", newline="", encoding="utf-8") as file:
    reader = csv.reader(file)
    headers = next(reader)
    data = list(reader)

hypothesis = None

for row in data:
    if row[-1].strip().lower() == "yes":
        attributes = row[:-1]
        if hypothesis is None:
            hypothesis = attributes.copy()
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = "?"

print("Attributes:", headers[:-1])
print("Final Hypothesis:", hypothesis)
```

Output



Result

Each attribute differs across the positive trip examples, so the final hypothesis has '?' for all attributes.

Problem 5: Choosing a Fitness Activity

Aim

To demonstrate FIND-S with a fitness-activity dataset in a separate CSV file and calculate the most specific hypothesis consistent with positive choices.

Procedure

21. Create fitness_data.csv with the activity attributes and target labels.
22. Keep the CSV and Python file in the same folder.
23. Read the data and initialize the hypothesis from the first ChooseActivity = Yes row.
24. Ignore No rows and replace differing values in positive examples with '?'.
25. Display the final hypothesis.

Dataset File / Input

File name: fitness_data.csv

The CSV file contains a header row followed by these training examples:

Goal	Intensity	Location	Duration	ChooseActivity
Cardio	Moderate	Outdoor	30min	Yes
Strength	High	Indoor	45min	Yes
Flexibility	Low	Indoor	20min	No
Cardio	High	Indoor	30min	Yes

Code

```
import csv

filename = "fitness_data.csv"

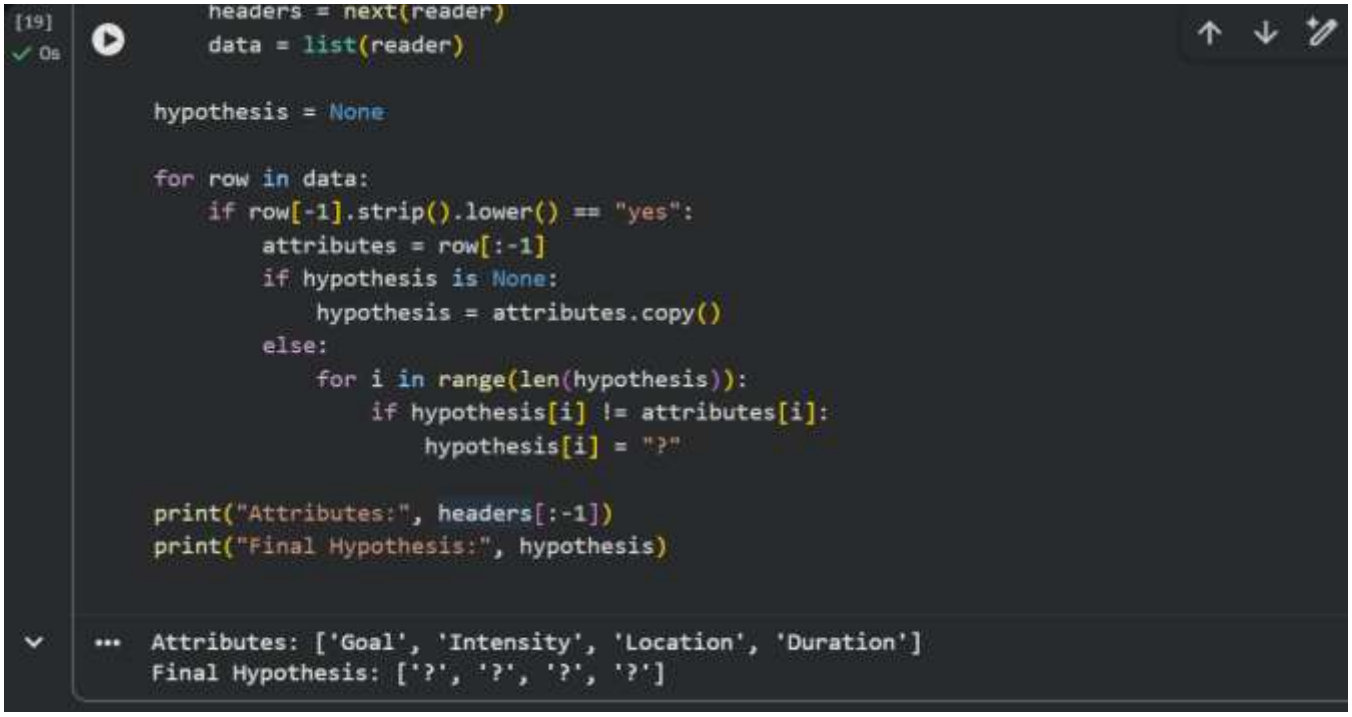
with open(filename, "r", newline="", encoding="utf-8") as file:
    reader = csv.reader(file)
    headers = next(reader)
    data = list(reader)

hypothesis = None

for row in data:
    if row[-1].strip().lower() == "yes":
        attributes = row[:-1]
        if hypothesis is None:
            hypothesis = attributes.copy()
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = "?"

print("Attributes:", headers[:-1])
print("Final Hypothesis:", hypothesis)
```

Output



```
[19]
✓ 0s
    headers = next(reader)
    data = list(reader)

hypothesis = None

for row in data:
    if row[-1].strip().lower() == "yes":
        attributes = row[:-1]
        if hypothesis is None:
            hypothesis = attributes.copy()
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = "?"

print("Attributes:", headers[:-1])
print("Final Hypothesis:", hypothesis)

...
Attributes: ['Goal', 'Intensity', 'Location', 'Duration']
Final Hypothesis: ['?', '?', '?', '?']
```

Result

All attributes vary across the positive choices, so the algorithm generalizes every attribute to '?'.